PRE-IDEATHON TEMPLATE

GENESIS IDEATHON

TEAM & DOMAIN

Introduce your team and the domain you have chosen

- Team Name – Ignite
- **Team Members**
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- Selected Domain – #8
- Problem Statement – Space Debris posing safety threats



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DOMAIN PROBLEM IN OUR WORDS

Explain the problem in your domain simply, in your own words

- **What exactly needs to be solved?**
  Excess Clutter of Space Debris causing multiple risks

- **Why is it important right now?**
  Space debris as of today has become a pressing issue due to neglect for decades. Many spacecrafts that are no longer functional are still orbiting the Earth's atmosphere have been causing a safety risk for new spacecrafts that are being sent to the orbit or out into space. These cause physical obstruction, as well as a major resource wastage. It also is a pollutant risk. A lot of the unmanaged debris ends up either staying in orbit or falls rashly onto Earth and due to the high friction in the mesosphere it either explodes or burns up in the atmosphere and that is never a safe option neither for the impact and economical effects nor the environment as some materials encased in these crafts are hazardous if exposed like the fuel and coolants.

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WHO IS AFFECTED?

Target users, stakeholders, beneficiaries

- **Primary users / target group**
  Government Space agencies and Defence Organisations(ISRO, NASA, DRDO…)
  Commercial satellites and satellite constellation operators Reliance JIO, TATA Sky…)
  Spacecraft Engineers and Aerospace manufacturers (Skyroot…)
  Space traffic management (upcoming STL)
- **Secondary stakeholders**
  Academic and scientific researchers
  Aviation and terrestrial safety Regulators
  Commercial launch service providers
- **Who benefits if the problem is solved?**
  Global Economy and public Society
  Space agencies and commercial satellite operators

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CURRENT SITUATION & CONTEXT

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Existing scenario, background and how things work today

Existing scenario and background

Space around Earth is getting crowded. In 2025 alone, over 300 launches added more than 4,000 new objects into orbit. Satellites help with phone networks, GPS, weather forecasts and more, but old and broken ones never really leave — they just keep circling, along with debris from past crashes and failed missions.

Relevant data

As per ESA's July 2026 report: 47,140 objects are tracked, 68,450 are estimated above 10 cm, 1.5 million fragments are between 1–10 cm, and 230 million are smaller than 1 cm. Debris below 600 km falls back in months, but between 600–2,500 km it can stay up for centuries.

Current workflow, step by step

Radars and telescopes spot an object → its orbit gets logged → systems check if it's getting close to another satellite → the operator is warned → if the satellite has fuel, it moves out of the way → the debris itself is never removed, so the same risk comes back again later.

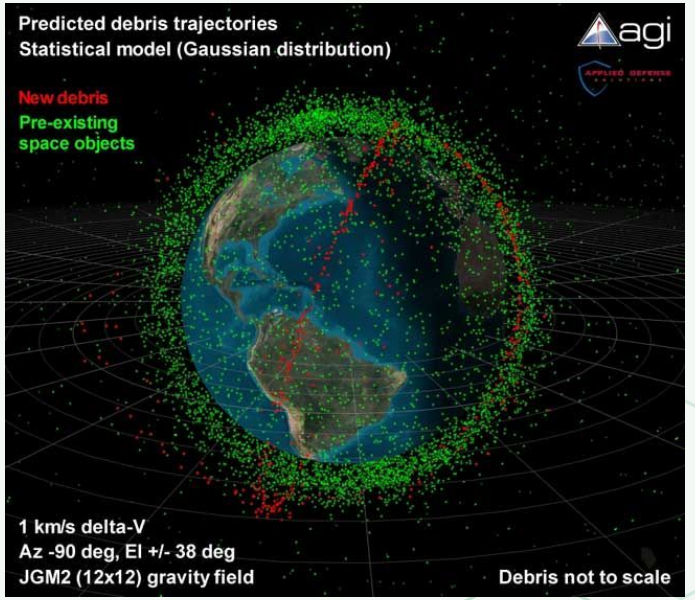


Image: Analytical Graphics, Inc. (AGI), Iridium 33–Cosmos 2251 collision debris simulation, STK/Systems Tool Kit.

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KEY CHALLENGES

Main difficulties, bottlenecks, root causes and consequences

Main difficulties and bottlenecks

We can only track objects bigger than 10 cm properly — millions of smaller fragments stay invisible, yet can still disable a satellite. Catching debris is also risky since most of it is tumbling with nothing to grab onto, so a failed capture can create even more pieces. Countries can't even remove each other's old debris without permission, and dead objects have no way to steer clear of anything on their own.

Root causes behind them

Older missions were never built to clean up after themselves. Leftover fuel in dead rocket bodies has caused explosions in orbit. The rise of mega-constellations has packed orbits faster than tracking could keep up. Past weapon tests (like China's in 2007 and Russia's in 2021) scattered thousands of fragments in seconds. Also, since objects move at around 7.5 km/s, shielding heavy enough to stop them would make satellites too heavy to launch.

Consequences of leaving them unsolved

One collision can trigger more collisions, which is what's called the Kessler syndrome. Left unchecked, this could cause global economic losses worth almost 2% of GDP, make satellite replacement extremely costly, and increase risk for crewed missions like the ISS, which already has to dodge debris regularly.

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EXISTING SOLUTIONS / APPROACHES

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What is already being done, and where it falls short

- **Existing methods, systems or alternatives**
  ADRAS-J (Japan): Identification of debris
  ADARAS-J2(Japan): Targeting and capturing the body ADRAS 1 imaged
  DFSM(Debris Free Space Mission): By ISRO, committed to have debris free for Indian Space Actors by 2030
- **Limitations of each**
  ADRAS-J(Japan): Only identified
  ADRAS-J2(Japan): It's a future endeavour and specified only Craft to be cleaned up
  DFSM(Debris Free Space Mission): Limited to Indian space actors
- **The gap they leave open (Key Challenges)**
  ADRAS-J(Japan): No action only information gathering
  ADARAS-J2(Japan): Only targeted and can only complete one item per round
  DFSM(Debris Free Space Mission): Not Much has been revealed so far and hence no identifiable gaps or advantages

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PROBLEM INSIGHTS & OPPORTUNITY AREAS

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Resources required to solve the problem

- **Hardware Requirements**
  Sensors(Temperature, Light, Pressure), GPS, Fuel, Solar Panels, Recycled material(i.e plastic), Controller chip, antenna, charging ports, Rechargeable big battery(TBD) connected with solar panels, smaller rechargeable batteries, charging dock, storage unit, Vacuum suction

- **Software Requirements**
  Tracking From Imaging, Locating Mothership, Fullness of ship, GPS for drop off and pick up. AI Imaging with human training, Temperature and pressure sensing, explosion probability and threat management

- **Other resources (Including Human Resources)**
  Field Professionals in Aerospace and research

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INITIAL DIRECTION & EXPECTED OUTCOME

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What Is Your Expected Outcome?

- What your team intends to do
  We intend to substantially decrease space debris, to improve safety in aerospace as well as recover potential financial losses.

- What you want to improve
  Quality of research and academics and future of spacecrafts

- What successful improvement should you achieve
  Efficiency with sustainability, saving prices, safer atmosphere,increased scope for aerospace research

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TEAM CONTRIBUTION

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Who will handle which part of the execution

Team Member	Role	Responsibility in Execution
Member 1 name	Role	What they will handle
Member 2 name	Role	What they will handle
Member 3 name	Role	What they will handle
Member 4 name	Role	What they will handle